

Academia is the cradle of new research, and teaching is its foundation. I teach because I enjoy it immensely—and because, over a decade of doing it in classrooms from Bangalore to Washington, teaching has repaid me every time with a deeper understanding of my own subject and a clearer voice for expressing it. What has changed since I first stood in front of a classroom is not the enthusiasm but the evidence: I now design and run the graduate and undergraduate spine of a data science program, build open tools and public course sites that outlast the semester, and carry students from their first lab to co-authorship on research papers.

As I learn, so I teach. During the early stage of my student life in India, I mostly resorted to self-teaching, owing to the scarcity of competent mathematics teachers. Those long, enervating episodes spent apprehending an elusive concept taught me what a good teacher is worth—and, in teaching myself, I kept discovering how I would teach the concept to someone else. Long before I had students, I taught imagined ones. The conviction stuck: *as I learn, so I teach*. It carried me from preparing students for competitive examinations to co-founding a small tutoring institute in Bangalore—six employees, undergraduate calculus, analysis, and linear algebra, and a modest profit—whose real yield was the certainty that I wanted a Ph.D. and a teaching career.

Training and experience. At *Tulane*, four years as a teaching assistant in statistics, calculus, and applied mathematics taught me to look through a student's eyes, and a summer as instructor of record for undergraduate statistics let me put that to work. At *UC Berkeley*, I taught graduate *Statistics for Data Science* for the MIDS program—a top-ranked online master's—for three years, developing curricula, materials, and teaching technology for a flipped, video-first format; my instructor ratings consistently sat above the department average. Since 2023, as faculty in the *Data Science Program at the George Washington University*, I have taught across the program's entire vertical: graduate *Introduction to Data Mining* (DATS 6103), *Algorithm Design* (DATS 6001), *Computer Science Foundations*, *Linear Algebra for Data Science*, and—new this fall—*Introduction to Data Science* (DATS 6101); undergraduate *Data Science for All* and the year-long *Data Science Capstone*, where teams take a real dataset through the full life cycle; and, as visiting faculty at NIT Sikkim, a graduate course in *topological data analysis*—from topological spaces to persistent homology—for computer science students. The span is deliberate: I can teach the mathematics as mathematics, the computing as computing, and the data science as the craft that joins them.

Philosophy: interaction, structure, and honest feedback. No unified method accounts for all students; persistent *interaction*—inside and outside the classroom—is what lets me meet the differences in their mathematical and cultural backgrounds. My classes blend lecture with *active learning* in proportions set by the course and the audience: theory-heavy courses stress logical reasoning and writing; applied courses are built around problem solving with real data. I structure every course as a weekly rhythm students can trust: a lab released on schedule, an online discussion thread I seed and tend, solutions published after the deadline, and formative feedback throughout—quizzes, polls, and office hours with an agenda. I collect anonymous mid-course feedback rather than waiting for final evaluations, and I invite colleagues to observe my classroom; from constructive evaluations to demeaning online reviews, I take every source seriously enough to locate the problem behind it.

The toolsmith's classroom. I am a coding hobbyist, and my courses show it. When GW's client-isolated campus wifi made it impossible for students to share code with the projector, I wrote *classroom-relay*, a small open-source websocket service that lets any student push code to the front of the room in one keystroke—a five-minute fix for a friction that had quietly taxed every class session. My courses live on *public websites* I build and maintain (smajhi.com/DATS-6101, smajhi.com/DATS-6103), open to anyone with no login, with discussion built in; my NIT Sikkim TDA course lives in a *public repository*. For

concepts that resist chalk—modes of stochastic convergence is my favorite example—I build *interactive visualizations* and publish them as tutorials at smajhi.com/tutorials. And my research group’s browser playgrounds let a newcomer run topological data analysis—with statistical certificates—without installing anything or paying for compute. The principle underneath all of it: remove every friction between a student and the idea, and make the removal available to everyone, not just my roster.

From classroom to co-authorship. The capstone course and my research program form a pipeline I tend deliberately. My recent papers carry student co-authors from undergraduate to doctoral—an undergraduate co-author on a fluid-dynamics paper now under journal review, master’s mentees guided into research, and a doctoral co-advisee whose thesis spans my Euler-characteristic program. Where they go next is the measure I care about: past mentees are now in graduate programs at Johns Hopkins, Zurich, and Montana State, at GW’s Regulatory Studies Center, and in industry—and one, a doctoral co-author, is a postdoctoral scholar at the Chennai Mathematical Institute. The open-source library my group builds (*Akriti*, on PyPI) is staffed by a student developer team with a real contribution process—onboarding documents, code review, and credit—because software engineering, like proof writing, is taught by apprenticeship.

What I can teach. Mathematics: *statistics and probability, linear algebra, analysis, topology, computational geometry, differential geometry*. Computing and data science: *algorithms and data structures, programming, data mining, machine learning, data science across the curriculum*, and specialized graduate courses in *topological data analysis* and *geometric and topological methods for machine learning*—the last two drawn directly from my research program, and field-tested at NIT Sikkim. I am equally at home designing a new course from scratch (I have done it five times in three years) and inheriting a large service course with an established syllabus.